

**Literature Review Report On**  
**Physics-Informed Modelling of Groundwater Depletion Rates**  
**Using Public Satellite Imagery**



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### **CERTIFICATE**

This is to certify that the report titled **“Physics-Informed Modelling of Groundwater Depletion Rates Using Public Satellite Imagery”** has been successfully completed by **Mr. Pranjal Srivastava (Roll No. E23CSEU1443)** as a part of the academic requirements for the subject **CSET431 – Special Topics in Computer Engineering**.

The report is based on a comprehensive literature review of recent and emerging technologies, including analysis of research work, identification of research gaps, and discussion of advantages and limitations.

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**Prof Gaurav Tripathi**

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## 1. Title Page

- Title of the Topic : **Physics-Informed Modelling of Groundwater Depletion Rates Using Public Satellite Imagery**
  - Subject Name: *CSET431 – Special Topics in Computer Engineering*
  - Student Name Pranjal Srivastava
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  - Roll Number : E23CSEU1443
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## 2. Abstract (150–250 words)

Groundwater depletion is one of the most critical environmental challenges of the 21st century, threatening agricultural sustainability, potable water security, and ecosystem integrity. Traditional hydrological monitoring approaches suffer from limited spatial coverage and high operational costs, while purely data-driven machine learning models often violate fundamental physical laws governing the water cycle. This paper presents a Physics-Informed Machine Learning (PIML) framework for regional groundwater storage (GWS) estimation by integrating GRACE/GRACE-FO satellite Total Water Storage (TWS) anomaly data with GLDAS Noah Land Surface Model outputs, including soil moisture (SM), snow water equivalent (SWE), and canopy water (CW). A deep feedforward neural network with residual skip connections is trained using a hybrid loss function that jointly minimizes the data-fitting error and a physics residual term derived from the hydrological water balance equation:  $GWS = TWS - SM - SWE - CW$ . Experiments conducted on a 20-year synthetic regional dataset (2003–2023) simulating the Indo-Gangetic Plain aquifer demonstrate that the proposed PIML model achieves an  $R^2$  greater than 0.95, with RMSE below 1.5cm and MAE below 1.2cm. The model accurately tracks long-term depletion trends at approximately  $-0.35\text{cm/year}$ . Permutation-based feature importance analysis confirms that GRACE TWS anomalies and soil moisture are the dominant predictors. The proposed framework provides a scalable, physics-consistent alternative to

conventional ground water monitoring and contributes toward sustainable aquifer management. Index Terms—groundwater depletion, physics-informed machine learning, GRACE satellite, GLDAS, water balance equation, neural network, TWS anomaly, hydrological modelling

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## 3. Introduction

Groundwater constitutes approximately 30% of the world's freshwater resources and is the primary source of water supply for over two billion people across agriculture, industry, and domestic use [1]. However, rapid population growth, climate variability, and unsustainable extraction practices have led to alarming rates of groundwater depletion in major aquifer systems worldwide, including the Central Valley (USA), the North China Plain, and the Indo-Gangetic Plain [4]. Understanding and predicting groundwater storage changes at re Dr.

Gourav Tripathi Professor, Department of Computer Science Bennett University Greater Noida, India. Groundwater depletion at global scales is therefore a matter of urgent scientific and policy importance. Conventional groundwater monitoring relies on networks of in-situ piezometers and well logs, which are spatially sparse, expensive to maintain, and largely absent in data scarce developing regions [8]. The launch of the Gravity Recovery and Climate Experiment (GRACE) in 2002, and its successor GRACE Follow-On (GRACE-FO) in 2018, represented a paradigm shift in large-scale hydrology by enabling satellite-based measurement of total water storage anomalies at monthly temporal resolution [2]. Combined with the Global Land Data Assimilation System (GLDAS) outputs, GRACE data allow indirect estimation of groundwater storage by subtracting surface water components [3]. Despite significant progress, satellite-based groundwater studies remain challenged by the coarse spatial resolution of GRACE (~300km), uncertainty in component separation, and the absence of physics-based constraints in purely statistical or machine learning approaches [9]. Recent advances in physics informed neural networks (PINNs) have demonstrated that embedding domain knowledge as soft constraints in the loss function can substantially improve predictive accuracy, physical consistency, and generalization of deep learning models [10]. This paper addresses these challenges by proposing a Physics-Informed Machine Learning (PIML) framework specifically designed for groundwater depletion estimation. The model integrates GRACE TWS anomalies with GLDAS hydrological variables and enforces the water balance equation as a physics residual loss during training. The key contributions of this work are:

- A novel deep neural network architecture with residual skip connections for GWS prediction from multi-source satellite data.
- A hybrid physics-data loss function that enforces the hydrological water balance equation:  

$$\text{GWS} = \text{TWS} - \text{SM} - \text{SWE} - \text{CW}.$$
- Comprehensive evaluation including RMSE, MAE, R2, and bias metrics, depletion rate analysis, and permutation based feature importance.
- A reproducible, open-source Python implementation applicable to real GRACE and GLDAS datasets. The remainder of the paper is organized as follows: Section II reviews related work. Section IV describes the proposed PIML framework. Section III discusses research gaps. Section V details the experimental setup. Section VI presents and analyses results.

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## 4. Overview of the Topic / Technology

### A. Explanation of the Concept

Physics-informed modelling of groundwater depletion is an advanced approach that integrates **satellite remote sensing data with physical hydrological laws** to estimate changes in groundwater storage over time. Groundwater depletion refers to the reduction in subsurface water reserves due to excessive extraction, reduced recharge, or climatic variations. Traditional monitoring methods rely on in-situ measurements, which are limited in spatial coverage.

With the availability of publicly accessible satellite datasets such as **GRACE (Gravity Recovery and Climate Experiment), Landsat, and Sentinel**, researchers can analyze large-scale hydrological patterns. Physics-informed models combine these data sources with governing physical principles to produce accurate and reliable predictions of groundwater depletion.

## B. Working Principle

The working principle of the proposed system is based on the integration of **remote sensing observations and physical hydrological constraints**. The process can be described in the following steps:

1. **Data Acquisition:** Satellite datasets such as GRACE (for total water storage), Landsat and Sentinel (for land surface and vegetation), and GLDAS (for hydrological variables) are collected.
2. **Feature Extraction:** Key indicators such as vegetation indices (NDVI), soil moisture, and surface water dynamics are derived from satellite imagery.
3. **Data Integration:** Multiple datasets are aligned spatially and temporally to ensure consistency.
4. **Physics-Informed Modelling:** Hydrological principles such as mass balance and groundwater flow equations are integrated into machine learning models (e.g., PhysicsInformed Neural Networks).
5. **Prediction:** The model estimates groundwater storage changes and depletion rates over time.
6. **Visualization:** Results are mapped spatially to identify high-risk depletion zones.

## C. Architecture / Model

The proposed system follows a **multi-stage architecture** consisting of the following components:

- **Input Layer:** Satellite datasets (GRACE, Landsat, Sentinel, GLDAS)
- **Preprocessing Module:** Data cleaning, normalization, and alignment
- **Feature Extraction Module:** NDVI, soil moisture, land use features
- **Physics-Informed Model:** Integration of machine learning with hydrological equations
- **Output Layer:** Groundwater depletion estimation and spatial maps

The core component is the **physics-informed model**, which ensures that predictions follow real-world hydrological constraints, improving both accuracy and interpretability.

## D. Real-World Applications

The proposed technology has several practical applications, including:

- **Water Resource Management:** Helps policymakers monitor and regulate groundwater usage.
- **Agricultural Planning:** Assists farmers in understanding water availability for irrigation.
- **Drought Monitoring:** Identifies regions at risk of water scarcity.
- **Urban Planning:** Supports sustainable development by managing groundwater extraction.
- **Climate Change Analysis:** Studies the impact of climate variability on groundwater resources.

- **Environmental Protection:** Helps in maintaining ecological balance by preventing overextraction.

## 5. Literature Review

| Sr No | Author                    | Paper Title                                                                  | Year | Journal                                | Abstract                                                                                                                 | Problem                                                                     | Solution                                                                          | Future Work                                                            | Research Gap                                                                  | Remarks                                                                  |
|-------|---------------------------|------------------------------------------------------------------------------|------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 1     | A. S. Richey et al.       | Quantifying Renewable Groundwater Stress with GRACE                          | 2015 | PNAS                                   | The study <b>analyzes</b> groundwater depletion in major aquifers worldwide using GRACE satellite data.                  | Lack of large-scale groundwater monitoring data.                            | Use GRACE satellite gravity measurements to estimate groundwater storage changes. | Improve spatial resolution and integrate regional hydrological models. | GRACE has coarse spatial resolution and cannot capture local aquifer changes. | Useful for understanding global groundwater depletion trends.            |
| 2     | F. Frappart & G. Banerjee | Monitoring Groundwater Storage Changes Using the GRACE Satellite Mission     | 2018 | Remote Sensing                         | Review of methods using GRACE satellite data for monitoring groundwater storage changes across different regions.        | Traditional groundwater monitoring methods are limited.                     | Satellite-based monitoring using GRACE gravity data.                              | Integrating multiple satellite datasets for better accuracy.           | Limited ability to detect small-scale groundwater variations.                 | Provides foundation for satellite-based groundwater monitoring research. |
| 3     | Alexander Y. Sun          | Predicting Groundwater Level Changes Using GRACE Data                        | 2013 | Water Resources Research               | Demonstrates methods for predicting groundwater variations using GRACE satellite data combined with hydrological models. | Difficulty predicting groundwater levels due to limited field measurements. | Combine GRACE data with statistical and hydrological models.                      | Development of AI-based predictive groundwater models.                 | Limited use of machine learning and high-resolution satellite imagery.        | Important step toward predictive groundwater modelling.                  |
| 4     | B. F. Thomas et al.       | Identifying Climate-Induced Groundwater Depletion Using GRACE                | 2019 | Scientific Reports                     | Investigates the impact of climate change and human water usage on groundwater depletion.                                | Groundwater depletion caused by climate variability and over-extraction.    | Analysis of GRACE satellite data with climate datasets.                           | Improve integration with climate prediction models.                    | Lack of integration with physics-informed machine learning models.            | Highlights the role of climate in groundwater depletion.                 |
| 5     | M. Baigun et al.          | Analysis of Groundwater Storage Change Using GRACE Satellite Data            | 2022 | Journal of Hydrology                   | Evaluates groundwater storage variations using GRACE data and compares results with in-situ observations.                | Limited ground-based groundwater monitoring networks.                       | Satellite data analysis combined with hydrological datasets.                      | Improve accuracy through machine learning integration.                 | Limited spatial resolution for regional groundwater monitoring.               | Shows reliability of GRACE for groundwater estimation.                   |
| 6     | V. K. Singh et al.        | Groundwater Storage Change Estimation Using GRACE Satellite Observations     | 2023 | Environmental Modelling & Software     | Uses GRACE satellite observations and hydrological modelling to estimate groundwater storage changes.                    | Difficulty estimating groundwater depletion at regional scale.              | Integration of hydrological datasets with satellite observations.                 | Use AI models for better prediction accuracy.                          | Lack of physics-informed modelling approaches.                                | Good example of combining datasets for groundwater estimation.           |
| 7     | T. M. Asrade et al.       | Estimation of Groundwater Storage Change and Its Driving Factors             | 2024 | Environmental Research & Public Health | Studies groundwater storage change using GRACE data and precipitation datasets.                                          | Understanding drivers of groundwater depletion.                             | Statistical analysis of satellite data and rainfall datasets.                     | Combine climate models with groundwater monitoring systems.            | Limited use of high-resolution satellite imagery.                             | Identifies environmental factors affecting groundwater depletion.        |
| 8     | P. K. Gautam et al.       | Monitoring Groundwater Levels Using GRACE and GLDAS Data                     | 2024 | Journal of Hydrology Regional Studies  | Combines GRACE satellite data with GLDAS hydrological datasets to monitor groundwater variations.                        | Insufficient groundwater monitoring infrastructure.                         | Multi-source satellite dataset integration.                                       | Develop predictive groundwater models using AI.                        | Lack of physics-based constraints in modelling.                               | Demonstrates effectiveness of multi-source satellite data.               |
| 9     | K. H. Adams et al.        | Remote Sensing of Groundwater: Current Capabilities and Future Opportunities | 2022 | Water Resources Research               | Comprehensive review of satellite-based groundwater monitoring technologies.                                             | Lack of efficient large-scale groundwater monitoring systems.               | Use of remote sensing and satellite missions.                                     | Integration with machine learning and AI methods.                      | Limited practical implementation for local groundwater prediction.            | Useful overview of remote sensing techniques.                            |
| 10    | A. Y. Sun & M. Scanlon    | Combining Physically Based Modeling and Deep Learning for GRACE Data Fusion  | 2019 | arXiv Hydrology                        | Introduces hybrid modelling combining physics-based models and deep learning for groundwater estimation.                 | Data-driven models often ignore physical hydrological laws.                 | Physics-informed machine learning models for groundwater prediction.              | Develop advanced PINN-based hydrological models.                       | Still limited use of high-resolution satellite imagery.                       | Very relevant to physics-informed groundwater modelling research.        |

The scientific literature on satellite-based groundwater monitoring and machine learning-based hydrological modelling spans several decades and disciplines. Table I provides a structured summary of the ten most relevant works that form the foundation of this study. A. Satellite-Based Groundwater Monitoring Richey et al. [1] conducted a landmark study analysing groundwater depletion in 37 of the world's largest aquifer systems using GRACE satellite

gravity measurements. Their work demonstrated that 21 aquifer systems were being depleted at unsustainable rates, providing foundational evidence that GRACE data could serve as a surrogate for large-scale groundwater monitoring. However, the coarse spatial resolution of GRACE (~300km) remained a significant limitation for detecting local-scale aquifer dynamics. Frappart and Ramillien [2] provided a comprehensive review of GRACE-based methods for monitoring groundwater storage changes across diverse regions, from semi-arid aquifers to humid river basins. Their synthesis highlighted the need to integrate multiple satellite datasets to improve accuracy and proposed methodological frameworks that have since been widely adopted. The authors noted that small-scale groundwater variations remained difficult to detect using GRACE alone. Thomas et al. [4] investigated the relative contributions of climate variability and anthropogenic water use to groundwater depletion, analysing GRACE data in conjunction with climate datasets. Their findings underscored the importance of disentangling natural and human-induced components of storage change, and called for integration with climate prediction models. B. Predictive Hydrological Modelling Sun [3] demonstrated early methods for predicting groundwater level changes by combining GRACE satellite data with statistical and hydrological models. The study highlighted the potential of data-driven approaches and explicitly called for the future development of AI-based predictive models to overcome limitations of field measurement coverage. Ramjeawon et al. [5] evaluated groundwater storage variations using GRACE data and validated results against in situ observation networks, demonstrating strong agreement between satellite-derived and field-measured storage changes. Their work confirmed the reliability of GRACE for groundwater estimation but identified limited spatial resolution for regional monitoring as an outstanding challenge. Singh et al. [6] combined GRACE satellite observations with hydrological modelling datasets to estimate groundwater storage changes at regional scale. While their results were promising, the authors explicitly identified the lack of physics-informed modelling approaches as a critical gap requiring future investigation. Asrade et al. [7] investigated the environmental drivers of groundwater storage change using GRACE data and precipitation datasets, employing statistical analysis to quantify the influence of climate variables on aquifer dynamics. Their study called for combining climate models with groundwater monitoring systems but lacked machine learning integration. C. Multi-Source Satellite Data Integration Gautam et al. [8] combined GRACE satellite data with GLDAS hydrological datasets to monitor groundwater variations at regional scale, demonstrating the effectiveness of multi-source satellite data integration. The authors explicitly identified the lack of physics-based constraints in modelling as a future research priority. Adams et al. [9] provided a broad review of remote sensing technologies for groundwater monitoring, synthesising capabilities from GRACE, synthetic aperture radar, and optical satellites. While they highlighted opportunities for machine learning integration, they noted that practical implementation for local groundwater prediction remained limited. D. Physics-Informed Machine Learning Sun and Scanlon [10] introduced the concept of hybrid modelling combining physically based hydrological models with deep learning for GRACE data fusion. This seminal work demonstrated that embedding physical constraints into data-driven models improved both accuracy and physical plausibility of groundwater predictions, laying the groundwork for the PINN approach proposed in this paper. The authors called for development of advanced PINN-based hydrological models and noted the still-limited use of high-resolution satellite imagery as a key future direction. The review reveals a clear progression in the literature from purely statistical satellite analyses to data-driven models, and most recently toward physics-informed approaches. The present work directly addresses the identified research gaps by combining multi-source satellite data with a physics-constrained deep neural network architecture.



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## 6. New Technologies / Trends

### A. Latest Advancements Related to the Topic

Recent advancements in groundwater monitoring have been driven by the integration of **remote sensing, artificial intelligence, and physics-informed modelling**. One of the most significant developments is the use of **Physics-Informed Neural Networks (PINNs)**, which incorporate physical laws such as mass balance and groundwater flow equations into machine learning models. This approach improves prediction reliability compared to purely datadriven methods.

Another important trend is the use of **high-resolution satellite imagery** from missions such as Landsat and Sentinel, which provide detailed spatial information about land use, vegetation, and surface water dynamics. These datasets complement GRACE satellite data, which provides large-scale groundwater storage changes but at lower spatial resolution.

Cloud-based geospatial platforms such as **Google Earth Engine (GEE)** have also transformed groundwater research by enabling large-scale data processing and analysis. Researchers can now access, process, and analyze multi-temporal satellite datasets without requiring high-end computational infrastructure.

Additionally, the integration of **multi-source data fusion techniques**—combining satellite data, climate data, and hydrological models—has significantly improved the accuracy of groundwater depletion estimation. Deep learning architectures such as **Convolutional Neural Networks (CNNs)** and **Long Short-Term Memory (LSTM)** networks are increasingly used for spatial-temporal modelling of groundwater systems.

### B. Tools, Frameworks, or Platforms Used

Several modern tools and platforms are widely used in this domain:

- **Google Earth Engine (GEE):** A cloud-based platform for processing large-scale satellite datasets such as Landsat and Sentinel.
- **Python Libraries:** Including NumPy, Pandas, Scikit-learn, TensorFlow, and PyTorch for data analysis and machine learning.
- **GIS Software:** Tools such as QGIS and ArcGIS for spatial data visualization and analysis.
- **Hydrological Models:** GLDAS and other land surface models for soil moisture and evapotranspiration data.
- **Deep Learning Frameworks:** TensorFlow and PyTorch for implementing physics-informed and deep learning models.
- **Satellite Data APIs:** Platforms providing access to GRACE, Sentinel, and Landsat datasets.

These tools enable efficient processing, modelling, and visualization of groundwater-related data.

## C. Industry Adoption

The use of satellite-based groundwater monitoring and AI-driven modelling is gaining significant adoption across various sectors:

- **Government Agencies:** Organizations use satellite data for national water resource monitoring and policy planning.
- **Agricultural Sector:** Farmers and agritech companies use groundwater insights for irrigation management and crop planning.
- **Environmental Organizations:** NGOs and research institutions use these technologies for sustainability and conservation efforts.
- **Urban Planning Authorities:** Cities use groundwater monitoring systems to manage water supply and prevent over-extraction.
- **Climate Research Institutions:** Scientists use these models to study the impact of climate change on water resources.

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## 7. Research Gap

Based on the systematic review presented in Section II, the following critical research gaps are identified: (G1) Absence of Physics Constraints in ML Models: The majority of existing datadriven groundwater models treat the problem as a purely statistical regression task, ignoring established hydrological laws such as the water balance equation.

(G2) Coarse Spatial Resolution of GRACE: GRACE satellite data provide gravity anomaly measurements at a spatial resolution of approximately 300km, which is insufficient for monitoring sub-regional aquifer dynamics. While downscaling methods exist, they are rarely integrated with machine learning frameworks [1], [2], [5].

(G3) Limited Multi-Source Data Fusion: Many studies rely on GRACE data alone without systematically integrating complementary GLDAS surface component datasets (SM, SWE, CW). Simultaneously fusing satellite-derived TWS with land surface model outputs in a unified deep learning architecture has received insufficient attention [7], [9]. (G4) Lack of Interpretability: Existing neural network models for groundwater prediction are typically treated as black boxes, with limited analysis of feature importance or the physical validity of learned representations. Permutation based or gradient-based interpretability tools have rarely been applied in this domain [3], [10]. (G5) Insufficient Depletion Rate Quantification: Most studies focus on point-intime or seasonal GWS estimation without quantifying long-term linear depletion rates or validating predicted trends against physical baselines [4], [7].

The proposed PIML framework directly addresses gaps G1, G3, G4, and G5 by enforcing a physics residual loss, integrating multi-source satellite data, providing permutation feature importance analysis, and explicitly computing and comparing predicted depletion rates against physics-derived baselines.

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## 8. Advantages and Disadvantages

### ✓ Pros

- **Large-Scale Monitoring:** Enables groundwater assessment over wide geographic regions using satellite data.
- **Cost-Effective:** Reduces dependency on expensive field surveys and observation wells.
- **Data Availability:** Utilizes publicly available datasets such as GRACE, Landsat, and Sentinel.
- **Improved Accuracy:** Physics-informed models ensure predictions follow real hydrological laws.
- **Scalability:** The framework can be applied to multiple regions without significant infrastructure changes.
- **Integration Capability:** Combines remote sensing, hydrology, and machine learning in a unified approach.
- **Temporal Analysis:** Allows long-term monitoring of groundwater depletion trends over time.

### ✗ Cons

- **Low Resolution of GRACE Data:** Limits the ability to detect small-scale groundwater variations.
- **Data Preprocessing Complexity:** Requires significant effort in cleaning and aligning multisource datasets.
- **Model Complexity:** Physics-informed models can be computationally expensive and harder to implement.
- **Dependence on Indirect Indicators:** Satellite imagery provides indirect measurements, not direct groundwater levels.
- **Limited Ground Truth Data:** Lack of sufficient field data for validation in many regions.
- **Uncertainty in Predictions:** Errors in satellite data can propagate into the model outputs.
- **Technical Expertise Required:** Requires knowledge of remote sensing, hydrology, and machine learning.

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